$$y[n] = \frac{3x[n-1] + x[n-2]}{x[n-3]}$$

1) causality	✓
2) memory-less	X
3) linearity	X
4) time invarience	✓
5) stability	✓

3) homogeneity: $x'[n] = \alpha x[n]$

$$y'[n] = \frac{3x'[n-1] + x'[n-2]}{x'[n-3]} = \frac{3\alpha x[n-1] + \alpha x[n-2]}{\alpha x[n-3]} = y[n] = \alpha y[n]$$

X

additivity: $x'[n] = x_1[n] + x_2[n]$

$$y'[n] = \dots \neq y_1[n] + y_2[n] = \dots$$ X

4) $x'[n] = x[n+d]$

$$y'[n] = \frac{3x'[n-1] + x'[n-2]}{x'[n-3]} = \frac{3x[n+d-1] + x[n+d-2]}{x[n+d-3]} = y[n+d]$$

5) $x[n-1] < M_1 < \infty$

$x[n-2] < M_2 < \infty$

$x[n-3] < M_3 < \infty$

$$y[n] = \frac{3M_1 + M_2}{M_3} = 4 \qquad (M_1 = M_2 = M_3)$$

$y[n] = x[n] \cdot e^{\frac{n}{4}}$

1) causality	✓
2) memory-less	✓
3) linearity	✓
4) time invarience	✗
5) stability	✓

3) $x'[n] = \alpha x[n]$

$y'[n] = x'[n] \cdot e^{\frac{n}{4}} = \alpha x[n] \cdot e^{\frac{n}{4}} = \alpha y[n]$ homo ✓

$x'[n] = x_1[n] + x_2[n]$

$y'[n] = x'[n] e^{\frac{n}{4}} = (x_1[n] + x_2[n]) e^{\frac{n}{4}} = y_1[n] + y_2[n]$ add ✓

4) $x'[n] = x[n+d]$

$y'[n] = x'[n] \quad e^{\frac{n}{4}} = x[n+d] e^{\frac{n}{4}} \neq y[n+d] = x[n+d] \cdot e^{\frac{n+d}{4}}$

5) $x[n] < M_1$

$e^{\frac{n}{4}} \longrightarrow \infty$ (you have to check if the expression is bounded)

$y[n] < M_1 \cdot M_2 < \infty$

irgum bargum

y_n

$$Y[n] = \frac{1}{2}\left(X[2n] + X[2n+1]\right)$$

1) causality,	×
2) memory-less,	×
3) linearity,	✓
4) time invariance,	×
5) stability	✓

1) $n < 2n$ if $n > 0$

2) $n > 2n$ if $n < 0$

3) homogeneity:

$$X'[n] = \alpha X[n]$$

$$Y'[n] = \frac{1}{2}\left(\alpha X[2n] + \alpha X[2n+1]\right) = \alpha Y[n] \checkmark$$

additivity:

$$X'[n] = X_1[n] + X_2[n]$$

$$Y'[n] = \frac{1}{2}\left(X'[2n] + X'[2n+1]\right) =$$

$$= \frac{1}{2}\left(X_1[2n] + X_2[2n] + X_1[2n+1] + X_2[2n+1]\right) =$$

$$= \frac{1}{2}\left(X_1[2n] + X_1[2n+1]\right) + \frac{1}{2}\left(X_2[2n] + X_2[2n+1]\right)$$

$$= Y_1[n] + Y_2[n] \checkmark$$

4) $X'[n] = X[n+d]$

$$Y'[n] = \frac{1}{2}\left(X'[2n] + X'[2n+1]\right)$$

$$= \frac{1}{2}\left(X[2n+2d] + X[2n+2d+1]\right)$$

$$\neq Y[n+d] = \frac{1}{2}\left(X[2n+d] + X[2n+d+1]\right)$$

5) $X[2n] < M_1 < \infty$

$X[2n+1] \leq M_2 < \infty$

$M_1 + M_2 < \infty$

$Y[n] < M_1 + M_2 < \infty$